

Developer Mock Test Results

Test ID:	a0db736b-2689-42...	Student ID:	7040
Test Type:	Developer Assessment	Date:	2026-01-22 10:37:02
Overall Score:	0/25 (0.0%)	Performance:	Needs Improvement
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	0/10	0.0%	Needs Work
MCQ/Theory	0/10	0.0%	Needs Work
Coding	0/5	0.0%	Needs Work

Detailed Question Review

APTITUDE SECTION (0/10 - 0.0%)

Q1. Two people, A and B, can do a piece of work in 10 days and 15 days respectively. How many days will it take for A and B to do the work together?

Your Answer: 8 days

Correct Answer: 6 days

When A and B work together, their combined rate of work is the sum of their individual rates, so we add the fractions $\frac{1}{10}$ and $\frac{1}{15}$ to get $\frac{1}{6}$, meaning they can complete the work together in 6 days. It's likely that the mistake was in calculating the combined rate, perhaps adding the days instead of the rates or not finding a common denominator. Don't worry, it's an easy mistake to make, and with practice, you'll become more comfortable with these types of problems and get the correct answer, which is 6 days.

Q2. A mixture is made of two liquids A and B in the ratio 2:3. If the quantity of liquid A is 24 liters, what is the quantity of liquid B?

Your Answer: SKIPPED

Correct Answer: 36 liters

The correct answer is 36 liters because the ratio of liquid A to liquid B is 2:3, so if the quantity of liquid A is 24 liters, we can find the quantity of liquid B by multiplying 24 by $\frac{3}{2}$, which equals 36. If you skipped this question or got it wrong, you might have misunderstood how to apply the ratio to find the quantity of liquid B, but don't worry, it's an easy concept to learn and practice. Remember to always use the ratio to set up a proportion and solve for the unknown quantity, and you'll become more confident in your problem-solving skills.

Q3. The average of 10 numbers is 15. If one of the numbers is 20, what is the sum of the remaining 9 numbers?

Your Answer: SKIPPED

Correct Answer: 130

■ The correct answer is 130 because the average of 10 numbers being 15 means the total sum of the numbers is $10 * 15 = 150$, and if one of the numbers is 20, then the sum of the remaining 9 numbers is $150 - 20 = 130$. If you skipped this question, you might have been unsure about how to apply the concept of averages to solve for the sum of the remaining numbers. Don't worry, it's a great opportunity to practice and learn - remember that averages can be used to find total sums, and from there, you can solve for unknowns like the sum of the remaining numbers.

■ **Q4. There are five houses, each painted a different colour: blue, green, red, white, and yellow. Each house is occupied by a person of a different nationa...**

Your Answer: __SKIPPED__

Correct Answer: Blue

■ The correct answer is Blue because the Canadian lives in the first house, and since the American lives in the red house, the Canadian cannot live in the red house, and by process of elimination, the Canadian's house must be blue. If you chose a different answer, you might have misinterpreted the clues or missed the connection between the Canadian's house being the first one, which helps narrow down the options. Don't worry, it's all part of the learning process, and with practice, you'll become more comfortable with deductive reasoning and solving puzzles like this one.

■ **Q5. What is the next number: 3, 6, 9, 12, 15, ?**

Your Answer: __SKIPPED__

Correct Answer: 18

■ The correct answer is 18 because the pattern in the sequence is adding 3 to the previous number, so the next number would be $15 + 3 = 18$. If you chose a different answer, you might have misidentified the pattern or added a different number, but that's okay, it's all part of the learning process. Keep practicing and you'll become more comfortable recognizing patterns in sequences, which is an important math skill.

■ **Q6. What is the next number: 2, 6, 12, 20, 30, ?**

Your Answer: __SKIPPED__

Correct Answer: 42

■ The correct answer is 42 because the pattern in the sequence is formed by adding 4, 6, 8, 10, and so on, increasing the increment by 2 each time. This pattern suggests that the user likely understood the sequence but may have miscounted the increment or applied it incorrectly, leading to an incorrect answer. By recognizing and applying the pattern of increasing increments, you can improve your skills in identifying and completing sequences like this one.

■ **Q7. If price increases from Rs.200 to Rs.250, what is the percentage increase?**

Your Answer: __SKIPPED__

Correct Answer: 25%

■ To find the percentage increase, we calculate the difference in price, which is $\text{Rs.}250 - \text{Rs.}200 = \text{Rs.}50$, and then divide it by the original price, giving us $50/200 = 0.25$, or 25%. This is why the correct answer is 25%. If you got a different answer, you might have made a calculation error, such as incorrectly finding the difference or dividing by the wrong number, but don't worry, practicing with more examples will help you master this concept.

■ **Q8. A shopkeeper buys 100 items at Rs.10 each and sells 50 items at Rs.15 each. What is the percentage profit?**

Your Answer: __SKIPPED__

Correct Answer: 25%

■ The correct answer is 25% because the shopkeeper's cost price for 50 items is Rs.500 ($100 \text{ items} * \text{Rs.}10 * 0.5$) and the selling price for 50 items is Rs.750 ($50 \text{ items} * \text{Rs.}15$), resulting in a profit of Rs.250, which is 25% of the cost price. If you got it wrong, you might have calculated the profit percentage based on the total cost price of 100 items or misinterpreted the selling price of the items. Don't worry, it's all part of the learning process, and with practice, you'll become more comfortable with calculating profit percentages.

■ **Q9. A pipe can fill a tank in 6 hours. Another pipe can empty the tank in 8 hours. How long will it take to fill the tank if both pipes are open?**

Your Answer: __SKIPPED__

Correct Answer: 24 hours

■ The correct answer is 24 hours because when both pipes are open, the net rate of filling the tank is the difference between the rates of the two pipes, which is $1/6 - 1/8 = 1/24$ of the tank per hour. If you skipped this question or got it wrong, you might have forgotten to consider the combined effect of both pipes or miscalculated the net rate, but don't worry, it's a

great opportunity to practice and improve your problem-solving skills. By breaking down the problem step by step, you can develop a stronger understanding of rates and ratios, and become more confident in your abilities.

■ **Q10. A mixture is made of two liquids A and B in the ratio 3:5. If the quantity of liquid A is 36 liters, what is the quantity of liquid B?**

Your Answer: __SKIPPED__

Correct Answer: 60 liters

■ The correct answer is 60 liters because the ratio of liquid A to liquid B is 3:5, and if liquid A is 36 liters, we can find the quantity of liquid B by first finding the multiplier used to get from 3 to 36, which is 12, and then multiplying 5 by 12 to get 60. If you skipped this question or got it wrong, you might have misunderstood how to apply the ratio to find the quantity of liquid B, but don't worry, it's an easy concept to practice and master. Keep practicing and you'll become more confident in solving ratio problems like this one.

■ MCQ SECTION (0/10 - 0.0%)

■ **Q1. What is the purpose of using locks in Python threading?**

Your Answer: __SKIPPED__

Correct Answer: To prevent data corruption and ensure thread safety

■ The correct answer is right because locks in Python threading are used to synchronize access to shared resources, preventing multiple threads from modifying data simultaneously, which can lead to data corruption. If the user skipped this question, they might not have understood the importance of thread safety, which is crucial in multithreaded programming to avoid unexpected behavior. By using locks, developers can ensure that their program runs reliably and consistently, making it an essential concept to grasp in concurrent programming.

■ **Q2. What is the first step in the thread lifecycle?**

Your Answer: __SKIPPED__

Correct Answer: Import the threading module

■ The correct answer is to import the threading module because this is the initial step that allows you to work with threads in your program, providing access to the necessary classes and functions. If you skipped this question or chose a different option, you might have misunderstood the sequence of steps involved in creating and managing threads, possibly thinking that defining a function or starting the thread comes first. Don't worry, it's an easy mistake to make, and recognizing the importance of importing the necessary module is a great first step in learning about thread lifecycles.

■ **Q3. What is the purpose of the Global Interpreter Lock (GIL) in Python?**

Your Answer:

__SKIPPED__

Correct Answer:

To prevent multiple threads from executing Python bytecode concurrently

■ The correct answer is right because the Global Interpreter Lock (GIL) in Python is a mechanism that prevents multiple threads from executing Python bytecode concurrently, which helps to prevent issues with memory management and data corruption. If the user was wrong, they might have misunderstood the purpose of the GIL, thinking it allows for concurrent execution, when in fact it does the opposite to ensure thread safety. Don't worry if you got it wrong, it's a great opportunity to learn more about Python's internals and how to work effectively with threads in your programs.

■ **Q4. What is the correct step to read data from a file in Python?**

Your Answer:

__SKIPPED__

Correct Answer:

Open the file using the open() function, read data using read(), close the file

■ The correct answer is right because to read data from a file in Python, you need to open the file using the open() function, then use the read() method to retrieve the data, and finally close the file to free up system resources. If the user skipped the question, they might have been unsure about the correct method to read data, possibly confusing it with writing or

appending data to a file. Don't worry, it's an easy concept to learn, and with practice, you'll become proficient in working with files in Python.

■ Q5. What is the correct order of steps to execute a thread in Python?

Your Answer:

__SKIPPED__

Correct Answer:

Import the threading module, define a function to run in a thread, create a Thread object, start the thread

■ The correct answer is the right choice because it follows the logical sequence of steps to execute a thread in Python: first, you need to import the necessary module, then define the function that the thread will run, next create a Thread object with that function, and finally start the thread. If you skipped the question or chose a different option, you might have gotten the order of steps mixed up, which is an easy mistake to make, but don't worry, it's all part of the learning process. By practicing and reviewing the correct sequence, you'll become more comfortable and confident in working with threads in Python.

■ Q6. What is a best practice for using multithreading in Python?

Your Answer:

__SKIPPED__

Correct Answer:

Use multithreading for I/O-bound tasks

■ In Python, multithreading is most effective for I/O-bound tasks, such as reading or writing to files, networks, or databases, because it allows other threads to run while waiting for I/O operations to complete. Using multithreading for CPU-bound tasks is less effective due to the Global Interpreter Lock (GIL), which prevents multiple threads from executing Python bytecodes at once. If you chose option A, you might be misunderstanding the impact of the GIL on CPU-bound tasks, but don't worry, it's a common misconception, and recognizing the distinction between I/O-bound and CPU-bound tasks is a great step in improving your multithreading skills.

■ Q7. Why is synchronization required in multithreading?

Your Answer: __SKIPPED__

Correct Answer: To prevent data corruption and ensure data integrity

■ Synchronization is required in multithreading to prevent data corruption and ensure data integrity because when multiple threads access shared data, they can interfere with each other, leading to inconsistent or incorrect results. The correct answer is right because synchronization mechanisms, such as locks, help to coordinate thread access to shared data, preventing concurrent modifications that can cause data corruption. If the user was wrong, they might have misunderstood the primary purpose of synchronization, focusing instead on secondary benefits like performance or error handling, but with practice and review, they can develop a deeper understanding of multithreading concepts.

■ Q8. What is the first step in reading and writing files in Python?

Your Answer:

__SKIPPED__

Correct Answer:

Open the file using the open() function

■ The correct answer is to open the file using the open() function because this is the initial step that allows you to access and manipulate the file in Python. If you skipped this question or chose a different option, you might have mistakenly thought that reading or writing the file was the first step, but without opening the file, these actions cannot be performed. Don't worry, it's an easy concept to grasp, and with practice, you'll become more comfortable with the sequence of steps for working with files in Python.

■ Q9. What is the primary benefit of using multithreading in Python?

Your Answer: __SKIPPED__

Correct Answer: Increased responsiveness and performance in I/O-bound applications

■ The correct answer is right because multithreading in Python allows your program to perform other tasks while waiting for input/output operations to complete, making it ideal for I/O-bound applications like web scraping or network requests. If you skipped this question, you might have been unsure about the benefits of multithreading, but don't worry, it's a common area of confusion - just remember that multithreading shines in situations where your program spends a lot of time waiting for external inputs. By understanding this concept, you'll be able to write more efficient and responsive Python programs, so keep practicing and you'll get the hang of it.

■ Q10. What is the purpose of using try and except blocks in Python file handling?

Your Answer:

__SKIPPED__

Correct Answer:

To handle errors and exceptions during file operations

■ The correct answer is right because try and except blocks in Python file handling are used to catch and manage errors that may occur during file operations, such as a file not being found or permission being denied. If the user skipped this question, they may not have understood the importance of error handling in file operations, which is crucial to prevent program crashes and ensure a smooth user experience. By using try and except blocks, developers can write more robust and reliable code, and it's an essential concept to grasp for anyone working with files in Python.

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a Python program to download multiple files concurrently using threads. ****Input:**** A list of file URLs (e.g., `["http://example.com/file1.txt", ...`

Your Answer:

dfgh

Correct Answer:

```
import requests
import threading

def download_file(url):
    filename = url.split("/")[-1]
    response = requests.get(url)
    with open(filename, 'wb') as file:
        file.write(response.content)
    print(f"Downloaded {filename}")

def main():
    urls = ["http://example.com/file1.txt", "http://example.com/file2.txt",
"http://example.com/file3.txt"]
    threads = []

    for url in urls:
        thread = threading.Thread(target=download_file, args=(url,))
        threads.append(thread)
        thread.start()

    for thread in threads:
        thread.join()

    print("All files downloaded")

if __name__ == "__main__":
    main()
```

■ It looks like the student's code is incomplete, as they haven't provided any actual code, which is likely due to a misunderstanding of the problem or a lack of knowledge about concurrent programming in Python. The correct code works because it utilizes the `threading` module to create multiple threads that can download files concurrently, and the `requests`

library to handle the HTTP requests, allowing the program to efficiently download multiple files at the same time. By using threads and joining them at the end, the program ensures that all files are downloaded before printing the success message, making it a great example of how to achieve concurrent file downloads in Python.

■ **Q2. Write a Python program to count the number of lines in a given text file. ****Input:**** The name of the text file (e.g., "example.txt") ****Output:**** The n...**

Your Answer:

__SKIPPED__

Correct Answer:

```
def count_lines_in_file(filename):
    try:
        with open(filename, 'r') as file:
            lines = file.readlines()
            return len(lines)
    except FileNotFoundError:
        print(f"File '{filename}' not found.")
        return None

filename = input("Enter the filename: ")
line_count = count_lines_in_file(filename)
if line_count is not None:
    print(f"The number of lines in '{filename}' is: {line_count}")
```

■ It seems like the student's code was not provided, but I can still guide you on what might have gone wrong. The issue might be that the student's code didn't handle potential errors, such as the file not being found, or didn't properly read the file. The correct code works because it uses a `try-except` block to handle the `FileNotFoundError` and reads the file line by line using `readlines()`, then returns the count of lines using `len()`, making it a robust and reliable solution.

■ **Q3. Write a Python function to calculate the average grade of a student given their scores in three subjects: math, science, and English. ****Input:**** A dic...**

Your Answer:

__SKIPPED__

Correct Answer:

```
def calculate_average_grade(grades):
    return sum(grades.values()) / len(grades)

# Example usage:
grades = {"math": 85, "science": 90, "english": 78}
average_grade = round(calculate_average_grade(grades), 2)
print("Average grade:", average_grade)
```

■ It seems like the student's code was not provided, but that's okay. If they had issues, it might be due to incorrect calculation or handling of the dictionary input. The correct code works because it utilizes the built-in `sum()` function to add up all the scores in the dictionary and then divides by the number of subjects, obtained using `len(grades)`, to calculate the average grade, providing an accurate result when rounded to two decimal places.

■ **Q4. Write a Python class to manage bank account transactions. The class should have methods to deposit, withdraw, and check the balance. ****Input:**** Initia...**

Your Answer:

__SKIPPED__

Correct Answer:

```
class BankAccount:
    def __init__(self, initial_balance):
        self.balance = initial_balance

    def deposit(self, amount):
        self.balance += amount
```

```

def withdraw(self, amount):
    if amount > self.balance:
        print("Insufficient balance")
    else:
        self.balance -= amount

def check_balance(self):
    return self.balance

def main():
    initial_balance = 100
    transactions = [("deposit", 100), ("withdraw", 50), ("deposit", 200)]

    account = BankAccount(initial_balance)

    for transaction in transactions:
        if transaction[0] == "deposit":
            account.deposit(transaction[1])
        elif transaction[0] == "withdraw":
            account.withdraw(transaction[1])

    print("Final balance:", account.check_balance())

if __name__ == "__main__":
    main()

```

■ It appears that the student's code was not provided, but I can still offer guidance on what might have gone wrong. The issue in their code might be related to incorrect implementation of the class methods or not properly handling the transactions. The correct code works because it properly initializes a `BankAccount` class with an initial balance, then iterates through each transaction, using conditional statements to determine whether to deposit or withdraw the specified amount, ensuring accurate updates to the account balance.

■ **Q5. Write a Python program to crawl multiple web pages concurrently using threads. ****Input:**** A list of URLs (e.g., `["http://example.com/page1", "http://...**

Your Answer:

__FINAL_SUBMIT__

Correct Answer:

```

import requests
import threading

def crawl_page(url):
    try:
        response = requests.get(url)
        print(f"Crawled {url} with status code {response.status_code}")
    except Exception as e:
        print(f"Error crawling {url}: {e}")

def main():
    urls = ["http://example.com/page1", "http://example.com/page2", "http://example.com/page3"]
    threads = []

    for url in urls:
        thread = threading.Thread(target=crawl_page, args=(url,))
        threads.append(thread)
        thread.start()

    for thread in threads:

```

```
        thread.join()

    print("All pages crawled")

if __name__ == "__main__":
    main()
```

■ The issue with the student's code is that it's missing the actual implementation, as indicated by the `__FINAL_SUBMIT__` placeholder, which suggests that they haven't provided the necessary code to solve the problem. In contrast, the correct code works because it utilizes the `threading` module to create and manage threads that concurrently crawl the web pages, while the `requests` library handles the HTTP requests to retrieve the page content. By using threads and handling potential exceptions, the correct code provides a robust and efficient solution to crawl multiple web pages concurrently.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.